Aayush Mishra

Efficient and Reliable AI

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Efficient (post) training, inference and architectures | AI Safety | Interpretability | Misalignment Oversight & Control

EDUCATION

Johns Hopkins University Ph.D., Computer Science — Jan 2022 – present
M.S.E., Computer Science | GPA 4.0/4.0 (transcript) — Dec 2023

IIT Mandi B.Tech., Computer Science & Engineering, minor in Management — Aug 2015 – Jun 2019
GPA 8.88/10 (transcript)

EXPERIENCE

MATS x OpenAI AI Safety Research Fellow (11.0) Berkeley, CA — Sep 2026 – present

- Reliable activation probes for misalignment monitoring in LLMs.
- Steerability and Interpretability of looped reasoning models.

Qualcomm AI Research Intern Amsterdam, NL — May – Sep 2026

- Made Looped LMs (Ouro) truly dynamic: a best-available KV-caching strategy plus a difficulty-aware exit objective.
- *Cut FLOPs and KV-cache memory by up to 30% at full-depth quality*. Filed 3 invention disclosures.

Adobe AI Research Intern College Park, MD — May – Aug 2024

- Co-developed EigenLoRA, which recycles trained LoRA adapters into a shared principal subspace: *new tasks train up to 100× fewer parameters and multi-adapter inference uses ∼18× less memory*, for on-device LLM and diffusion models.

Microsoft Data & Applied Scientist, Bing Shopping Hyderabad, India — Aug 2021 – Jan 2022

- Built a query-to-product-category classifier over ∼18k classes and a relevance metric to improve offer ranking.

Siemens Research Engineer Bengaluru, India — Jul 2019 – Aug 2021

- Developed a CNN explainability and compression algorithm for edge deployment; Filed 3 invention disclosures.
- Trained an RL agent to surface edge-case safety violations of self-driving agents in simulation.
- Internships (2017–18): semantic code search with LSA; NLP classifier for Siemens' internal social network.

SELECTED PUBLICATIONS

Enabling Dynamic Computation in Looped LMs 2026
Aayush Mishra, et al. | Preprint | paper | thread

Steered LLM Activations are Non-Surjective 2026
Aayush Mishra, Daniel Khashabi, Anqi Liu | Sci4DL & Re-Align @ ICLR | paper | code | thread

IA2: Alignment with ICL Activations Improves Supervised Fine-Tuning 2026
Aayush Mishra, Daniel Khashabi, Anqi Liu | ICLR | paper | code | thread

Genomic Next-Token Predictors are In-Context Learners 2026
Nathan Breslow, *Aayush Mishra*, et al. | TMLR; **Oral** @ MLGenX (ICLR) | paper | code | thread

ODD: Overlap-aware Estimation of Model Performance under Distribution Shift 2025
Aayush Mishra, Anqi Liu | UAI | paper | code | thread

EigenLoRA: Recycle Trained Adapters for Resource-Efficient Adaptation and Inference 2024
*Aayush Mishra**, Prakhar Kaushik*, et al. | Preprint | paper

Do Pretrained Transformers Learn In-Context by Gradient Descent? 2024
*Aayush Mishra**, Lingfeng Shen*, Daniel Khashabi | ICML **Oral** | paper | talk

SKILLS & SERVICE

Tools & Methods Python, C++, PyTorch, Hugging Face, Transformers, TRL, vLLM; pre-training, post-training LLMs, LoRA/PEFT, distillation, reinforcement learning; steering/probing, interpretability, adaptive computation, KV-cache efficiency

Reviewer ICML (Silver Reviewer), ICLR, NeurIPS, UAI, TMLR